# When Prediction Is the Point

## Gov 51: Data Analysis and Politics

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Week 6
March 3, 2026

# Act I: Same Tool, New Purpose

# Thursday We Used Regression for Causality

| | **Broockman/Kalla 2016** |
|---|---|
| **Design** | Randomized experiment |
| **X** | Binary: treated vs. control |
| **Key parameter** | $\hat{\beta}_1 = 5.94$ (treatment effect) |
| **Covariates** | Bought precision, not different answers |

We cared about $\hat{\beta}_1$

# Today We Use Regression for Prediction

Same OLS machinery:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \cdots$$

| **Causality** | **Prediction** |
|---|---|
| What is $\hat{\beta}_1$? | How close is $\hat{Y}_i$ to $Y_i$? |
| One coefficient matters | All coefficients serve $\hat{Y}$ |
| Covariates $\rightarrow$ precision | Covariates $\rightarrow$ accuracy |

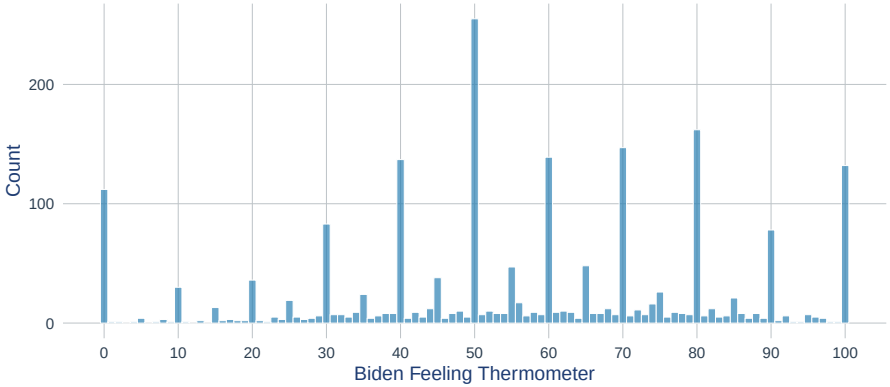
# Our Data: A National Political Survey

**Cooperative Election Study (2020-style)**

| Variable | Description | Range |
|---|---|---|
| `ftbiden` | Biden feeling thermometer | 0–100 |
| `pid7` | Party ID (1 = Strong Dem ... 7 = Strong Rep) | 1–7 |
| `age` | Age in years | 18–90 |
| `female` | 1 if female, 0 if male | 0/1 |
| `educ` | Education level | 1–6 |

$n = 2{,}000$ respondents

# The Outcome: Biden Feeling Thermometer




Spikes at 0, 50, 100 and multiples of 5/10 — survey heaping

# Can Party ID Predict How People Feel About Biden?

**Question:** If I know someone's party ID (1 = Strong Dem … 7 = Strong Rep), can I predict their Biden feeling thermometer (0–100)?

**Model:**

$$Y_i = \beta_0 + \beta_1 \cdot \text{pid7}_i + \varepsilon_i$$

$Y_i$: Biden thermometer for person $i$ $\beta_1$: how much $Y$ changes per unit of pid7

**In R:**

```
fit1 <- lm(ftbiden ~ pid7, data = d)
summary(fit1)
```

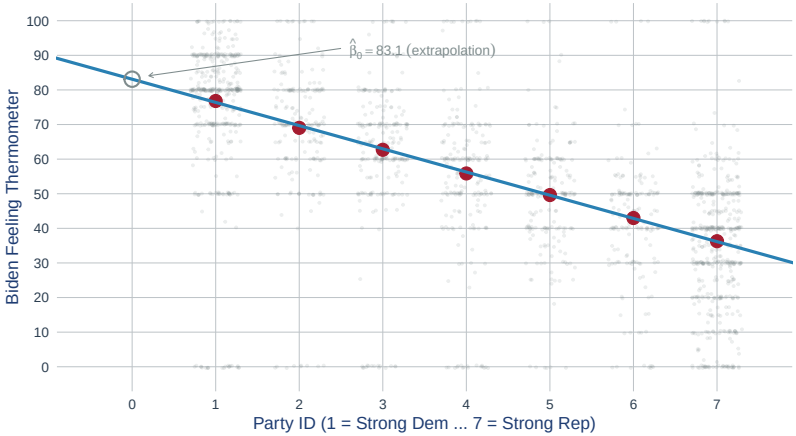
# One Predictor Already Explains 36% of the Variation

| | Estimate | Std. Error | t value | p-value |
|---|---|---|---|---|
| (Intercept) | 83.10 | 1.33 | 62.59 | $< 0.001$ |
| pid7 | $-6.71$ | 0.29 | $-23.22$ | $< 0.001$ |

$\hat{\beta}_0 = 83.10$: predicted thermometer for pid7 $= 0$ (extrapolation)
$\hat{\beta}_1 = -6.71$: each step toward Republican $\rightarrow$ 6.7 points colder
$R^2 = 0.355$: party ID explains 36% of variation

# Party ID Alone Predicts a Lot

$\hat{Y}_i = 83.10 - 6.71 \times \text{pid7}_i$ $R^2 = 0.355$



# Each Predictor Adds a Little More

| Model | Predictors | $R^2$ |
|---|---|---|
| pid7 only | 1 | 0.355 |
| + age | 2 | 0.382 |
| + female + educ | 4 | **0.404** |

$$R^2 = 1 - \frac{\text{SSR}}{\text{TSS}}$$

$R^2$ went up every time we added a predictor

# Reading an `lm()` Table

```
lm(ftbiden ~ pid7 + age + female + educ, data = d)
```

| | Estimate | Std. Error | t value | p-value |
|---|---|---|---|---|
| (Intercept) | 85.35 | 2.13 | 40.07 | $< 0.001$ |
| pid7 | $-6.66$ | 0.19 | $-34.25$ | $< 0.001$ |
| age | $-0.27$ | 0.03 | $-9.41$ | $< 0.001$ |
| female | 1.45 | 0.90 | 1.62 | 0.106 |
| educ | 2.81 | 0.34 | 8.35 | $< 0.001$ |

Each row tests $H_0 : \beta_j = 0$

# Statistical Significance $\neq$ Prediction Quality

| **p-value tells you** | **p-value does NOT tell you** |
|---|---|
| Whether $\hat{\beta}_j$ is distinguishable from zero | How well the model predicts $Y$ |
| With enough $n$, everything is "significant" | Whether $\hat{\beta}_j$ is practically large |

**Example:** `female` has $p = 0.106$ but the model predicts well

$R^2$ measures prediction quality, not p-values

# Act II: Letting the Data Bend

# OLS Fits the Best Line — But a Line May Not Be Best

**What $\hat{\beta}_1$ gives you:**

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 \cdot X_i \quad \Longrightarrow \quad \text{a straight line}$$

| What OLS does | What OLS doesn't do |
|---|---|
| Best *linear* fit | Best *overall* fit |
| $\hat{\beta}_1$ = slope (constant) | Allow the slope to change across $X$ |

**Solution:** add higher-order terms (polynomials) so $\hat{Y}$ can curve

# Sometimes the Average Doesn't Go Up in a Straight Line

| Predictor | Why the relationship curves |
|---|---|
| **Income** → turnout | Rises steeply at low income, flattens at high income |
| **Age** → participation | Peaks in middle age, declines among elderly |
| **Age** → Biden thermometer | Our data — let's check |

$\bar{Y}$ at each value of $X$ traces a curve, not a line

# Polynomials Let the Line Curve

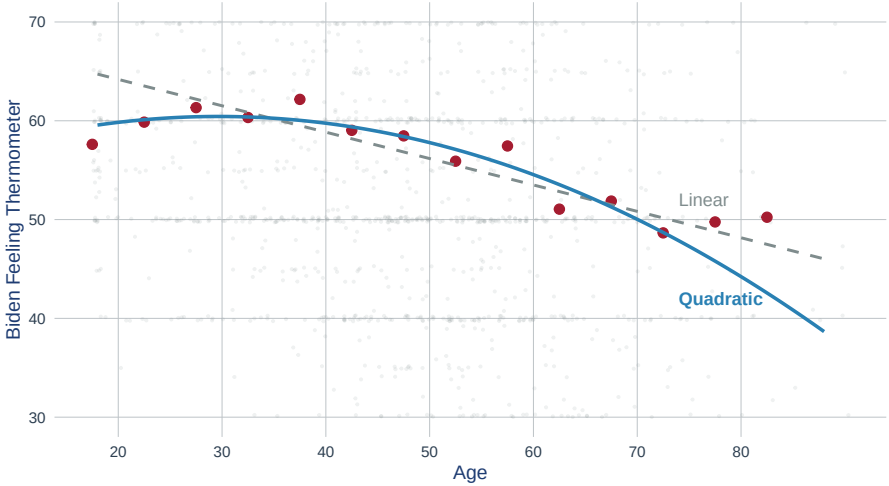
**Add age$^2$ as another predictor:**

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{age}_i + \hat{\beta}_2 \cdot \text{age}_i^2$$

| | Estimate | Std. Error |
|---|---|---|
| Intercept | 54.81 | 4.48 |
| age | 0.380 | 0.182 |
| age$^2$ | −0.0064 | 0.0018 |

Still OLS — just add `I(age^2)` as another column in `lm()`

# Age Has a Curved Relationship with Biden Approval



# Interactions: When the Effect Depends on Who You Are

**Does education predict Biden approval differently for men and women?**

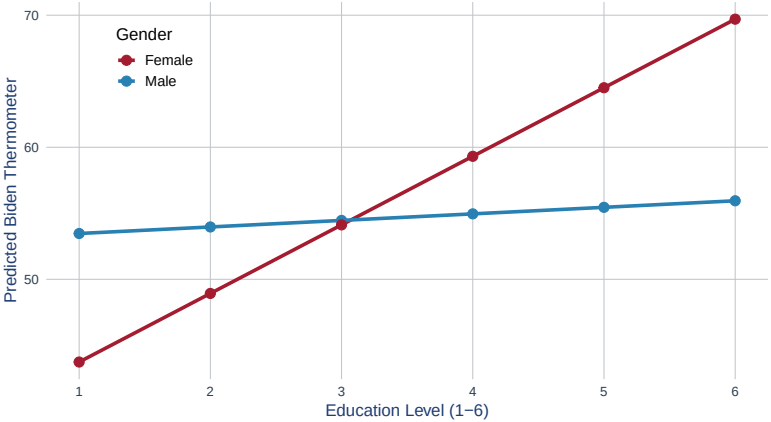
$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{educ}_i + \hat{\beta}_2 \cdot \text{female}_i + \hat{\beta}_3 \cdot (\text{educ}_i \times \text{female}_i)$

| | **Estimate** | **Std. Error** |
|---|---|---|
| educ | 0.50 | 0.63 |
| female | −14.43 | 3.27 |
| educ × female | **4.70** | 0.86 |

Male education slope: 0.50 points per level

Female education slope: $0.50 + 4.70 =$ **5.19** points per level

# The Interaction Changes the Slope




$\hat{\beta}_3$ measures how much the education slope differs by gender

# Three Types of Interactions

| Type | Example | What $\hat{\beta}_3$ measures |
|---|---|---|
| Binary × Continuous | female × educ | Slope difference |
| Binary × Binary | female × college | Difference in predicted means |
| Continuous × Continuous | age × pid7 | How one slope changes with the other |

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{1i} + \hat{\beta}_2 X_{2i} + \hat{\beta}_3 (X_{1i} \times X_{2i})$$

# Binary × Continuous: Plug In and Switch On

$$\hat{Y}_i = \underbrace{52.97}_{\hat{\beta}_0} + \underbrace{0.50}_{\hat{\beta}_1}\,\text{educ} \underbrace{-14.43}_{\hat{\beta}_2}\,\text{female} + \underbrace{4.70}_{\hat{\beta}_3}(\text{educ} \times \text{female})$$

| **Question** | **Set** | **Result** |
|---|---|---|
| Mean for males? | female $= 0$ | $\hat{Y} = 52.97 + 0.50 \cdot \text{educ}$ |
| Mean for females? | female $= 1$ | $\hat{Y} = 38.54 + 5.20 \cdot \text{educ}$ |
| Male, educ $= 3$? | plug in | $52.97 + 0.50(3) = \mathbf{54.5}$ |
| Female, educ $= 3$? | plug in | $38.54 + 5.20(3) = \mathbf{54.1}$ |
| F − M gap, educ $= 3$? | subtract | $54.1 - 54.5 = \mathbf{-0.4}$ |

**Female line:** $\hat{\beta}_0 + \hat{\beta}_2 = 38.54$ and $\hat{\beta}_1 + \hat{\beta}_3 = 5.20$

# Binary × Binary: Four Predicted Means

$$\hat{Y}_i = \underbrace{53.95}_{\hat{\beta}_0} \underbrace{-3.20}_{\hat{\beta}_1} \text{female} + \underbrace{1.43}_{\hat{\beta}_2} \text{college} + \underbrace{10.57}_{\hat{\beta}_3} (\text{female} \times \text{college})$$

| | **No College** | **College** |
|---|---|---|
| **Male** | $\hat{\beta}_0 = \mathbf{54.0}$ | $\hat{\beta}_0 + \hat{\beta}_2 = \mathbf{55.4}$ |
| **Female** | $\hat{\beta}_0 + \hat{\beta}_1 = \mathbf{50.7}$ | $\hat{\beta}_0 + \hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3 = \mathbf{62.7}$ |

Female − Male gap, no college: $50.7 - 54.0 = \mathbf{-3.2}$ $(= \hat{\beta}_1)$

Female − Male gap, college: $62.7 - 55.4 = \mathbf{+7.4}$ $(= \hat{\beta}_1 + \hat{\beta}_3)$

Difference of those differences: $7.4 - (-3.2) = \mathbf{+10.6}$ $(= \hat{\beta}_3)$

# Age Affects Democrats and Republicans Differently

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 \cdot \text{age}_i + \hat{\beta}_2 \cdot \text{pid7}_i + \hat{\beta}_3 \cdot (\text{age}_i \times \text{pid7}_i)$$

| | |
|---|---|
| $\hat{\beta}_3$ (age $\times$ pid7) | $-0.10$ |
| **Strong Democrat** (pid7 $= 1$): age slope | $+0.05$ |
| **Strong Republican** (pid7 $= 7$): age slope | $-0.56$ |

Older Democrats are slightly warmer toward Biden
Older Republicans are much colder

# $R^2$ Keeps Going Up

| Model | Predictors added | # Pred | $R^2$ |
|---|---|---|---|
| 1 | pid7 | 1 | 0.355 |
| 2 | + age | 2 | 0.382 |
| 3 | + female, educ | 4 | 0.404 |
| 4 | + age$^2$ | 5 | 0.407 |
| 5 | + pid7 $\times$ age, female $\times$ educ | 7 | **0.438** |

Every predictor increased $R^2$. Is this always a good thing?

# Act III: When More Is Worse

# A Parable: Memorizing vs. Learning

| | Practice Exam | Real Exam |
|---|---|---|
| **Student A:** memorized every problem | 100% | 60% |
| **Student B:** learned the patterns | 90% | 88% |

**Student A** = overfitting
Perfect on training data
Fails on new data

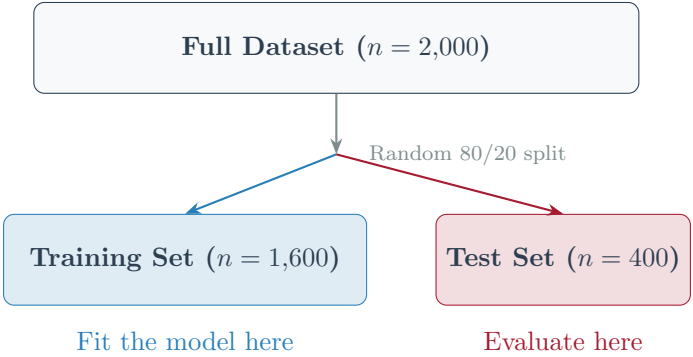
**Student B** = good model
Slightly worse on training data
Generalizes to new data

# In-Sample vs. Out-of-Sample

| In-sample | Out-of-sample |
|---|---|
| The data you fit on | New data the model has never seen |
| $R^2$ measures this | What actually matters |
| Always looks good | The honest test |

$R^2$ only tells you how well you fit the data you already have

# The Train/Test Split




**Rule:** Never let the model see the test data during fitting

# Four Lines of R

```r
# 1. Split
train_idx <- sample(1:nrow(d), size = 0.8 * nrow(d))
train <- d[train_idx, ]
test  <- d[-train_idx, ]

# 2. Fit on training data
fit <- lm(ftbiden ~ pid7 + age + female + educ,
          data = train)

# 3. Predict on test data
y_hat <- predict(fit, newdata = test)

# 4. Calculate test RMSE
sqrt(mean((test$ftbiden - y_hat)^2))
```

# RMSE: The Prediction Scorecard

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2}$$

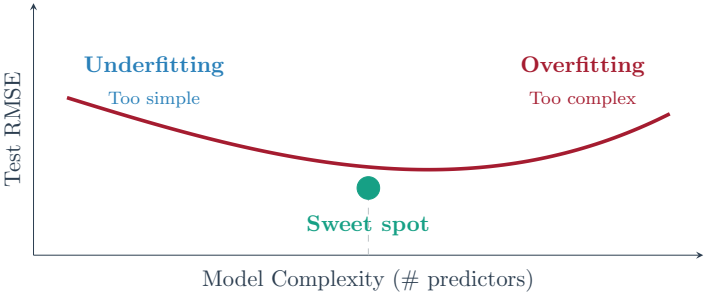
- In the units of $Y$ (thermometer points)
- Lower = better predictions
- **Train RMSE:** how well the model fits its own data
- **Test RMSE:** how well the model predicts *new* data

# The Experiment: Simple → Complex

| | Model | # Pred | Train RMSE | Test RMSE |
|---|---|---|---|---|
| 1 | pid7 only | 1 | 20.72 | 21.19 |
| 2 | + age | 2 | 20.19 | 21.15 |
| 3 | + female, educ | 4 | 19.82 | 20.80 |
| 4 | + $age^2$ | 5 | 19.78 | 20.70 |
| 5 | + interactions | 7 | 19.25 | **20.23** |
| 6 | 299 polynomial × interaction terms | 299 | 17.52 | **132.22** |

Train RMSE: 20.72 → 17.52 (always improving)
Test RMSE: 21.19 → 20.23 (improving) → 132.22 (catastrophe)

# Underfitting and Overfitting



# Why Overfitting Happens

| **Signal** | **Noise** |
|---|---|
| Real patterns in the population | Random variation in this sample |
| Repeats in new data | Does not repeat |

- Complex models fit both signal *and* noise
- With 299 predictors, the model memorized the training data
- Train RMSE = 17.52 (looks great)
- Test RMSE = 132.22 (memorization, not learning)

# What If We Keep Adding Polynomial Terms?

30 respondents, predicting Biden thermometer from age alone

**Degree 1:** $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \,\text{age}$

**Degree 3:** $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \,\text{age} + \hat{\beta}_2 \,\text{age}^2 + \hat{\beta}_3 \,\text{age}^3$

**Degree 7:** $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \,\text{age} + \cdots + \hat{\beta}_7 \,\text{age}^7$

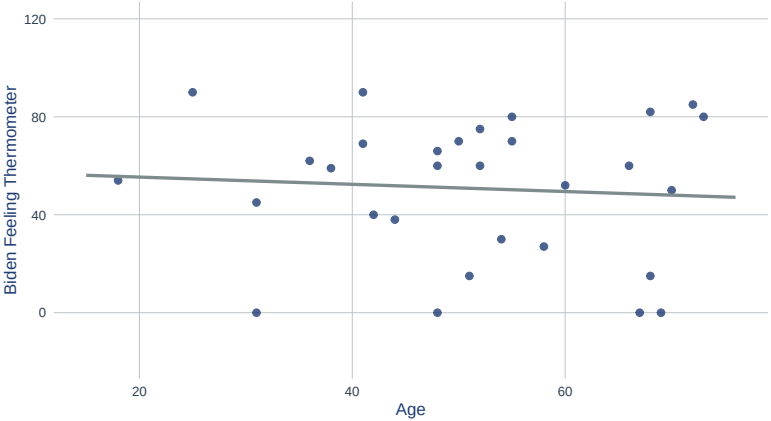
**Degree 20:** $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 \,\text{age} + \cdots + \hat{\beta}_{20} \,\text{age}^{20}$

More terms = more flexibility = better in-sample fit. But at what cost?

# Degree 1: A Straight Line

Train RMSE = 27.9

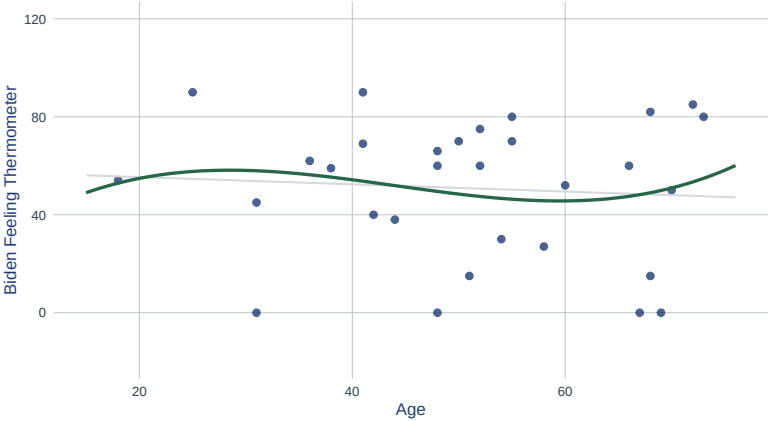
Test RMSE = 26.1



# Degree 3: A Little Curvature

Train RMSE = 27.7

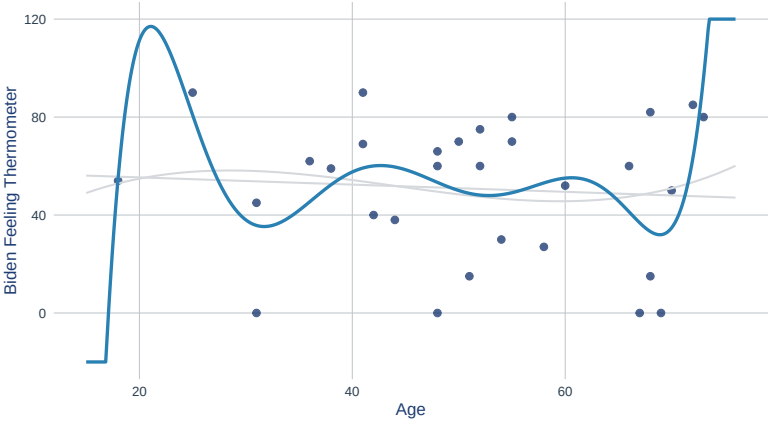
Test RMSE = 28.0



# Degree 7: Starting to Wiggle

Train RMSE = 24.5

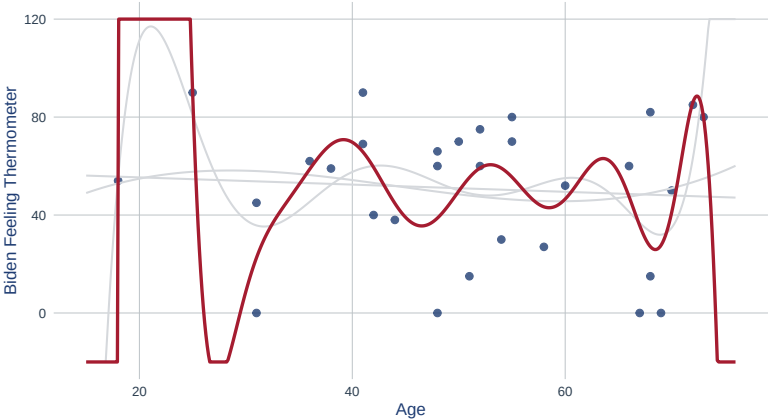
Test RMSE = 1,532



# Degree 20: Fitting Noise

Train RMSE = 22.0

Test RMSE = 506,682



# How Do People Find the Sweet Spot?

**Classic tools (you may see in textbooks):**

- **Adjusted $R^2$** — penalizes $R^2$ for adding predictors
- **AIC / BIC** — information criteria that trade fit vs. complexity
- **Mallow's $C_p$** — estimates out-of-sample prediction error

**The modern standard:**

- **Cross-validation** — split the data many times, average the test error

Next week: LASSO — uses cross-validation to automatically select predictors

**$R^2$ always goes up.**

**Test RMSE tells you the truth.**

**The field has tools to find the sweet spot — we'll learn the best one next week.**

# Coming Up

- **Thursday:** PS2 due (catching fraud with regression)
- **Next week:** LASSO — a method that automatically prevents overfitting
- **March 12:** Exam 1

*Questions?*